

JADAVPUR UNIVERSITY
B.E. ELECTRICAL ENGINEERING EXAMINATION - 2022
ELECTRICAL MACHINES - II (Part I)

Detailed Analytical Solutions provided by SR Sir

Question 1(a): Discuss the similarities between a transformer and an induction machine. Hence, explain why an induction machine is called a generalized transformer. [cite: 11, 12]

SR Sir's Thought Process

When approaching this question, we must trace the fundamental principles of operation. Both machines rely on Faraday's Law of Electromagnetic Induction. The transformer transfers power from one stationary circuit to another via a magnetic field. The induction motor transfers power from a stationary stator to a rotating rotor via a rotating magnetic field. The key distinction is the mechanical rotation, meaning the induction motor is effectively a transformer with a rotating secondary and an air gap. I will lay out the electrical, magnetic, and equivalent circuit similarities to provide a complete answer.

Answer:

Similarities between a Transformer and an Induction Machine:

- **Principle of Operation:** Both devices operate on the fundamental principle of Faraday's Law of Electromagnetic Induction. A time-varying flux links two separate windings, inducing an electromotive force (EMF) in both.
- **Power Transfer:** In both machines, power is transferred from the primary (stator) to the secondary (rotor) solely through mutual induction via a common magnetic field. There is no direct electrical connection between the primary and secondary circuits.
- **Flux Excitation:** Both machines require an exciting or magnetizing current to establish the working flux in the core (and air gap for the motor).
- **Equivalent Circuit:** The per-phase equivalent circuit of a 3-phase induction motor is identical in topological structure to that of a static transformer. Both have primary/stator impedance, a shunt magnetizing branch, and secondary/rotor impedance.

Why an Induction Machine is called a Generalized Transformer:

An induction machine is aptly termed a *generalized transformer* because its operation reduces exactly to that of a transformer under specific conditions. Let us analyze the frequency of the induced EMF in the secondary. For a transformer, the primary frequency f_1 and secondary frequency f_2 are strictly equal ($f_1 = f_2$). For an induction motor, the rotor frequency is given by:

$$f_2 = sf_1 \quad (1)$$

where s is the per-unit slip. If the rotor of the induction motor is locked or blocked so that it cannot rotate ($N_r = 0$), the slip $s = \frac{N_s - N_r}{N_s} = 1$. Under this standstill condition:

$$f_2 = (1)f_1 = f_1 \quad (2)$$

Thus, at standstill, the induction motor behaves exactly like a short-circuited transformer with an air gap. The rotational action simply introduces the slip parameter s , which scales the secondary voltage ($E_{2s} = sE_2$), frequency ($f_2 = sf_1$), and reactance ($X_{2s} = sX_2$). Because the static transformer is merely a special case of the induction machine (where $s = 1$ permanently), the induction motor is considered the more general case.

Question 1(b): What will be the direction of rotation of the rotor, if the primary winding is placed on the rotor of a poly-phase induction motor? [cite: 13]

SR Sir's Thought Process

This is a classic conceptual trick question. Standard induction motors have the primary (supply) on the stator. Here, the primary is on the rotor, and the stator is short-circuited (acting as the secondary). If we supply 3-phase AC to the rotor, it generates a rotating magnetic field (RMF) that rotates at synchronous speed N_s *relative to the rotor*. By Lenz's Law, the induced currents in the stator will try to oppose the relative motion. Since the stator is bolted to the ground and cannot move, the rotor itself must spin in the opposite direction to try and "catch up" and reduce the relative speed.

Answer:

If the primary winding is placed on the rotor and supplied with poly-phase currents, and the stator is short-circuited acting as the secondary, the rotor will rotate in the **opposite direction** to the direction of its own rotating magnetic field (RMF).

Reasoning: When the rotor is energized, it produces an RMF that revolves at synchronous speed (N_s) relative to the rotor structure itself. This moving flux cuts the stationary short-circuited stator conductors, inducing an EMF and consequently a current in them.

According to Lenz's Law, the electromagnetic torque developed will act in a direction that opposes the cause producing it. The cause here is the relative motion between the rotating flux and the stationary stator. To reduce this relative motion, the magnetic interaction attempts to drag the stator in the direction of the RMF. However, because the stator is physically fixed and cannot move, the reaction torque acts on the movable part—the rotor. Consequently, the rotor is forced to rotate in the direction **opposite** to the RMF. By doing so, the absolute speed of the RMF in space (and hence the relative speed cutting the stator) is reduced.

Question 1(c): Explain why the rotor of an induction motor can never attain synchronous speed while the rotor field with respect to the stator rotates with the synchronous speed. [cite: 14]

SR Sir's Thought Process

Two distinct concepts here: 1. Why $N_r \neq N_s$? Because if $N_r = N_s$, there is no flux cutting, no induced EMF, no torque. The motor would coast down due to friction until slip is restored. 2. Why does the rotor field rotate at N_s in space regardless of rotor speed? The rotor field moves at sN_s relative to the rotor. The rotor itself moves at N_r . Space speed = $N_r + sN_s = N_r + (N_s - N_r) = N_s$.

Answer:

Part 1: Why the rotor never attains synchronous speed For an induction motor to produce driving torque, a current must be induced in the rotor conductors. This induction only occurs if there is a relative velocity between the rotating magnetic field (RMF) produced by the stator and the rotor conductors. If the rotor were to somehow attain synchronous speed ($N_r = N_s$), the relative speed between the RMF and the rotor would be zero. Consequently:

- The flux cutting the rotor conductors would be zero.
- The induced rotor EMF and rotor current would be zero.
- The electromagnetic torque developed would become zero.

Without any driving torque, unavoidable mechanical losses (friction and windage) will immediately cause the rotor to decelerate. As it slows down ($N_r < N_s$), relative motion is restored, torque is generated, and the motor settles at a steady-state speed slightly below synchronous speed. Hence, it is physically impossible for motor action to occur at synchronous speed.

Part 2: Why the rotor field rotates at synchronous speed in space Although the rotor body rotates at speed N_r , the frequency of the currents induced in the rotor is the slip frequency, $f_2 = sf_1$. These slip-frequency currents create their own rotating magnetic field (the rotor field). The speed of this rotor field *with respect to the rotor body* is:

$$N_{\text{rotor-field}} = \frac{120f_2}{P} = \frac{120(sf_1)}{P} = sN_s \quad (3)$$

To find the speed of the rotor field with respect to the stationary stator (in space), we must add the mechanical speed of the rotor itself:

$$N_{\text{space}} = N_r + sN_s = N_r + \left(\frac{N_s - N_r}{N_s} \right) N_s = N_r + N_s - N_r = N_s \quad (4)$$

Thus, the rotor magnetic field always locks into synchronism and rotates at exactly N_s with respect to the stator, ensuring steady torque production.

Question 1(d): What do you mean by distributed winding? Find an expression for distribution factor. [cite: 15]

SR Sir's Thought Process

Instead of concentrating all coil turns of a phase in one slot per pole, we spread them over adjacent slots. This makes the EMF waveform more sinusoidal and improves cooling, though it slightly reduces the arithmetic sum of EMFs. The distribution factor K_d is the ratio of phasor sum to arithmetic sum of coil EMFs. I will draw the phasor diagram using TikZ to demonstrate the geometric derivation clearly.

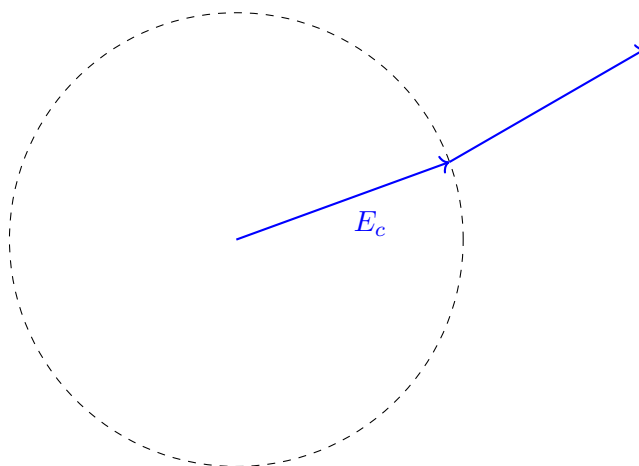
Answer:

Distributed Winding: In practical AC machines, all the conductors belonging to a single phase under one magnetic pole are not bunched together in a single slot. Instead, they are distributed across several adjacent slots. This type of winding is known as a distributed winding. The main advantages are: 1. It produces a more sinusoidal induced EMF waveform by reducing spatial harmonics. 2. It allows for better heat dissipation due to distributed copper losses. 3. It provides better mechanical strength and structural integrity to the core.

Expression for Distribution Factor (K_d): Because the coils of a phase are distributed in different slots, the EMFs induced in individual coils are not in phase with each other. They are displaced by an electrical angle γ (the slot angle). The Distribution Factor (or Breadth Factor), K_d , is defined as:

$$K_d = \frac{\text{Phasor sum of coil EMFs}}{\text{Arithmetic sum of coil EMFs}} \quad (5)$$

Let: m = number of slots per pole per phase γ = electrical angle between adjacent slots E_c = EMF induced in each coil



Note: Geometric derivation representation. Let the individual coil EMFs be represented by chords of a circle with center O and radius R .

Assume the individual coil EMF phasors $AB, BC, CD...$ each of magnitude E_c subtend an angle γ at the center O of the circumscribing circle.

$$E_c = AB = 2R \sin\left(\frac{\gamma}{2}\right) \quad (6)$$

The arithmetic sum of m such coils is:

$$E_{\text{arithmetic}} = mE_c = m \times 2R \sin\left(\frac{\gamma}{2}\right) \quad (7)$$

The resultant EMF is the phasor sum, which geometrically is the chord joining the start of the first phasor to the end of the last phasor. This chord subtends a total angle of $m\gamma$ at the center.

$$E_{phasor} = 2R \sin\left(\frac{m\gamma}{2}\right) \quad (8)$$

Therefore, the distribution factor K_d is:

$$K_d = \frac{E_{phasor}}{E_{arithmetic}} = \frac{2R \sin(m\gamma/2)}{m \times 2R \sin(\gamma/2)} \quad (9)$$

$$K_d = \frac{\sin\left(\frac{m\gamma}{2}\right)}{m \sin\left(\frac{\gamma}{2}\right)} \quad (10)$$

Question 1(e): The maximum torque of a three phase induction motor does not depend on the rotor resistance - Explain. [cite: 16]

SR Sir's Thought Process

This requires mathematical justification. I need to write the expression for electromagnetic torque T_e , find the condition for maximum torque by differentiating with respect to slip ($dT_e/ds = 0$), find the slip at max torque (s_m), and substitute s_m back into the torque equation to show that R_2 vanishes from the magnitude expression.

Answer:

To prove this, we must examine the fundamental torque equation of a three-phase induction motor. The electromagnetic torque developed is given by:

$$T_e = \frac{3}{\omega_s} \frac{V_1^2 \frac{R_2}{s}}{\left(R_1 + \frac{R_2}{s}\right)^2 + (X_1 + X_2)^2} \quad (11)$$

By applying the Maximum Power Transfer Theorem (or taking $\frac{dT_e}{ds} = 0$), maximum torque occurs when the variable rotor resistance referred to the stator ($\frac{R_2}{s}$) equals the magnitude of the equivalent impedance looking back into the stator. Assuming approximate equivalent circuit (or neglecting stator resistance R_1 for simplicity to highlight the core principle), the slip at which maximum torque occurs is:

$$s_{mT} = \frac{R_2}{\sqrt{R_1^2 + (X_1 + X_2)^2}} \approx \frac{R_2}{X_2} \quad (\text{if } R_1 \approx 0) \quad (12)$$

Notice that the slip at which maximum torque occurs is **directly proportional** to the rotor resistance R_2 .

Now, substitute s_{mT} back into the general torque equation:

$$T_{max} = \frac{3}{\omega_s} \frac{V_1^2 \left(\sqrt{R_1^2 + (X_1 + X_2)^2}\right)}{\left(R_1 + \sqrt{R_1^2 + (X_1 + X_2)^2}\right)^2 + (X_1 + X_2)^2} \quad (13)$$

Simplifying this yields the exact expression for maximum torque:

$$T_{max} = \frac{3V_1^2}{2\omega_s \left[R_1 + \sqrt{R_1^2 + (X_1 + X_2)^2}\right]} \quad (14)$$

Conclusion: In the final expression for T_{max} , the term R_2 is completely absent. Therefore, the magnitude of the maximum torque (pull-out torque) is entirely independent of the rotor resistance. However, increasing R_2 increases the slip s_{mT} at which this maximum torque occurs, thereby shifting the peak of the torque-slip curve towards starting ($s=1$).

Question 1(f): What is deep bar rotor induction motor? How it produces high starting torque? [cite: 17]

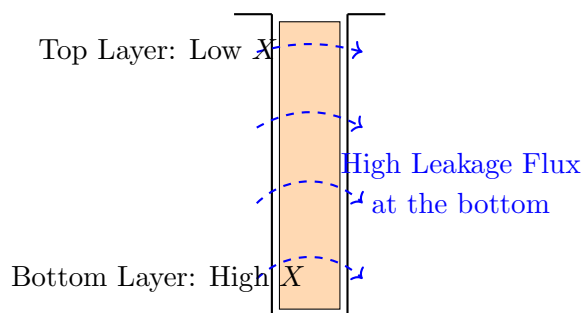
SR Sir's Thought Process

Deep bar rotors leverage the skin effect. The bars are narrow and very deep. At starting, $s = 1$, frequency is 50Hz . The bottom of the bar has high leakage inductance, so current is choked and pushed to the top. This reduces the effective cross-sectional area, increasing R_2 , yielding high starting torque. At running, $s \approx 0$, frequency is $1\text{-}2\text{Hz}$. Skin effect vanishes, current spreads uniformly, R_2 drops, increasing efficiency. Need a cross-sectional diagram.

Answer:

Deep Bar Rotor Induction Motor: A deep bar rotor is a special type of squirrel cage rotor construction designed to inherently provide high starting torque without sacrificing low-slip running efficiency. Instead of standard circular or square conductors, the rotor slots are punched very deep and narrow, and the copper or aluminum bars inserted into these slots have a high depth-to-width ratio.

Mechanism of Producing High Starting Torque (Skin Effect): The operation relies entirely on the **Skin Effect** governed by the rotor frequency. Consider a deep rotor bar divided into multiple horizontal layers. The bottom-most layer of the bar links with a much larger amount of leakage flux (flux that crosses the slot without crossing the air gap) compared to the topmost layer. Consequently, the leakage inductance (L) and leakage reactance ($X = 2\pi fL$) of the bottom layers are significantly higher than those of the top layers.



- **At Starting ($s = 1$):** The rotor frequency equals the supply frequency (e.g., 50 Hz). At this high frequency, the reactance of the lower portion of the bar is so high that the current is forced to flow almost entirely in the top portion of the bar (Skin Effect). This drastically reduces the effective cross-sectional area of the conductor, resulting in a high effective rotor resistance (R_2). Since starting torque $T_{st} \propto R_2$, the motor develops a high starting torque.
- **At Running ($s \approx 0.02$ to 0.05):** The rotor frequency is very low (e.g., 1 to 2.5 Hz). At such low frequencies, the leakage reactance is negligible ($X \approx 0$). The current distributes evenly across the entire depth of the bar. The full cross-sectional area is utilized, resulting in a low effective rotor resistance. This low resistance ensures low copper losses and high running efficiency.

Question 2(a): Prove that, a three phase balanced supply on a balanced three phase winding can produce a rotating magnetic field with constant magnitude and with synchronous speed. [cite: 19]

SR Sir's Thought Process

This is the fundamental proof of AC machinery. I need to define three sinusoidal currents displaced by 120 degrees in time. Then, define the spatial MMFs generated by three coils displaced by 120 degrees in space. Multiply them, apply trigonometric identities ($\cos A \cos B$), sum them up, and show that the resultant is a forward-traveling wave with magnitude $1.5F_m$.

Answer:

Let us consider a stator with a balanced 3-phase winding distributed in space, with the magnetic axes of phases A, B, and C displaced from each other by 120° (electrical). Let these windings be fed by a balanced 3-phase AC supply. The instantaneous currents are displaced in time by 120° :

$$i_A(t) = I_m \cos(\omega t) \quad (15)$$

$$i_B(t) = I_m \cos(\omega t - 120^\circ) \quad (16)$$

$$i_C(t) = I_m \cos(\omega t - 240^\circ) = I_m \cos(\omega t + 120^\circ) \quad (17)$$

Each phase current produces an alternating Magnetomotive Force (MMF) that is distributed sinusoidally along the air-gap space. Let θ be the spatial angle measured from the axis of phase A. The spatial MMFs produced by the individual phases are:

$$F_A(\theta, t) = N_{eff} i_A(t) \cos(\theta) = F_m \cos(\omega t) \cos(\theta) \quad (18)$$

$$F_B(\theta, t) = N_{eff} i_B(t) \cos(\theta - 120^\circ) = F_m \cos(\omega t - 120^\circ) \cos(\theta - 120^\circ) \quad (19)$$

$$F_C(\theta, t) = N_{eff} i_C(t) \cos(\theta - 240^\circ) = F_m \cos(\omega t + 120^\circ) \cos(\theta + 120^\circ) \quad (20)$$

where $F_m = N_{eff} I_m$ is the peak MMF of a single phase.

The total resultant MMF $F_{res}(\theta, t)$ in the air gap is the sum of the individual phase MMFs:

$$F_{res} = F_A + F_B + F_C \quad (21)$$

We apply the trigonometric identity: $\cos X \cos Y = \frac{1}{2} [\cos(X - Y) + \cos(X + Y)]$

For Phase A:

$$F_A = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta)] \quad (22)$$

For Phase B:

$$F_B = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta - 240^\circ)] \quad (23)$$

For Phase C:

$$F_C = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta + 240^\circ)] \quad (24)$$

Summing these three terms, we separate the forward-traveling terms ($\omega t - \theta$) and backward-traveling terms ($\omega t + \theta$):

$$\begin{aligned} F_{res} &= \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t - \theta) + \cos(\omega t - \theta)] \\ &\quad + \frac{F_m}{2} [\cos(\omega t + \theta) + \cos(\omega t + \theta - 240^\circ) + \cos(\omega t + \theta + 240^\circ)] \end{aligned} \quad (25)$$

Notice that the second bracket consists of three balanced co-phasal vectors separated by 120° (since $240^\circ \equiv -120^\circ$). The sum of any three balanced sinusoidal quantities is zero. Therefore, the backward-traveling wave cancels out entirely.

$$F_{res}(\theta, t) = \frac{3}{2}F_m \cos(\omega t - \theta) \quad (26)$$

Conclusion: The equation $F_{res} = 1.5F_m \cos(\omega t - \theta)$ represents a wave with a **constant magnitude** of $1.5F_m$. To find its velocity, we track a point of constant phase (e.g., the peak where $\omega t - \theta = 0$). This gives $\theta = \omega t$. Differentiating with respect to time gives the angular velocity:

$$\frac{d\theta}{dt} = \omega = 2\pi f \text{ (rad/sec)} \quad (27)$$

In mechanical speed, this corresponds to the synchronous speed $N_s = \frac{120f}{P}$. Thus, the field rotates at a constant synchronous speed.

Question 2(b): Derive an expression for torque developed in an induction motor. Draw the torque-speed characteristics. [cite: 20]

SR Sir's Thought Process

Torque derivation usually starts from Air Gap Power (P_{ag}). $P_{ag} = 3I_2'^2 \frac{R_2'}{s}$. Synchronous wattage concept dictates $T_{em} = \frac{P_{ag}}{\omega_s}$. I need to express I_2' in terms of V_1 using the equivalent circuit (Thevenin equivalent is rigorous, but approximate equivalent is standard for general derivation). I will use the standard exact equivalent circuit approximation. Then draw the curve using pgfplots highlighting starting, max torque, and full load regions.

Answer:

Derivation of Torque Equation: From the per-phase equivalent circuit of an induction motor referred to the stator, the rotor current I_2' is given by:

$$I_2' = \frac{V_1}{\sqrt{\left(R_1 + \frac{R_2'}{s}\right)^2 + (X_1 + X_2')^2}} \quad (28)$$

The power transferred across the air gap from stator to rotor (P_{ag}) is the power consumed in the rotor circuit resistance. For a 3-phase motor:

$$P_{ag} = 3(I_2')^2 \frac{R_2'}{s} \quad (29)$$

The electromagnetic torque developed (T_e) is fundamentally the air-gap power divided by the synchronous speed in mechanical radians per second ($\omega_s = \frac{2\pi N_s}{60}$):

$$T_e = \frac{P_{ag}}{\omega_s} = \frac{3}{\omega_s} (I_2')^2 \frac{R_2'}{s} \quad (30)$$

Substituting the expression for I_2' into the torque equation gives:

$$T_e = \frac{3}{\omega_s} \frac{V_1^2 \frac{R_2'}{s}}{\left(R_1 + \frac{R_2'}{s}\right)^2 + (X_1 + X_2')^2} \text{ N-m} \quad (31)$$

Torque-Speed Characteristics: By analyzing the derived equation, we can plot T_e against speed (or slip). 1. Low slip region ($s \approx 0$, near synchronous speed): (R_2'/s) is very large, so $T_e \approx ks$. The curve is linear. 2. High slip region ($s \approx 1$, starting): (R_2'/s) is small, so $T_e \approx k/s$. The curve is rectangular hyperbolic.

Question 3(a): Draw and explain the phasor diagram of a three phase induction motor and develop the equivalent circuit. [cite: 22]

SR Sir's Thought Process

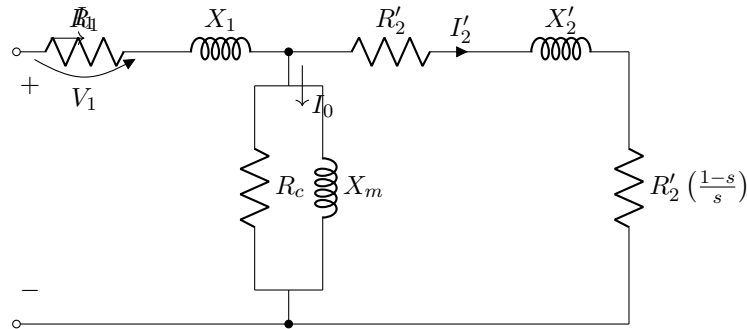
The development follows directly from a generalized transformer model. The stator is the primary, the rotor is a short-circuited secondary. 1. Draw the per-phase equivalent circuit utilizing 'circuitikz'. Show R_1, X_1 , magnetizing branch R_c, X_m , and referred rotor branch $R'_2/s, X'_2$. 2. Draw the phasor diagram. Taking Φ_m as reference, E_1 lags by 90. I_m and I_c make up I_0 . I'_2 is in phase with E_2 but lags by rotor impedance angle. $I_1 = I_0 + (-I'_2)$. $V_1 = -E_1 + I_1 Z_1$.

Answer:

Development of Equivalent Circuit: An induction motor behaves like a transformer with a rotating, short-circuited secondary. Let E_1 be the stator induced EMF per phase, and E_2 be the rotor induced EMF per phase at standstill ($s = 1$). At any slip s , the rotor induced EMF is sE_2 , and the rotor leakage reactance is sX_2 . The rotor current is:

$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} = \frac{E_2}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} \quad (32)$$

This mathematical division by s is physically equivalent to a stationary circuit where the voltage is E_2 and the resistance is $\frac{R_2}{s}$. We can further split this resistance into the actual copper resistance R_2 and a fictitious load resistance $R_2 \left(\frac{1-s}{s}\right)$ which represents the mechanical power developed. Referring these secondary parameters to the primary (stator) using the effective turns ratio, we obtain the per-phase exact equivalent circuit:



Phasor Diagram Explanation:

- **Reference:** The mutual air-gap flux Φ_m is taken as the reference phasor.
- **Induced EMFs:** The flux induces EMF E_1 in the stator and E'_2 (referred) in the rotor. Both lag the flux Φ_m by 90° according to Faraday's law. The component to overcome this in the primary is $-E_1$.
- **No-Load Current (I_0):** Comprises the magnetizing current I_m (in phase with Φ_m) and core-loss current I_c (in phase with $-E_1$).
- **Rotor Current (I'_2):** Flows due to E'_2 . It lags E'_2 by the rotor impedance angle $\theta_2 = \tan^{-1} \left(\frac{sX'_2}{R'_2} \right)$.

- **Stator Current (I_1):** The vector sum of no-load current I_0 and the load component of primary current $-I'_2$ (which directly opposes rotor current).
- **Supply Voltage (V_1):** The vector sum of $-E_1$ and the stator impedance drop $I_1(R_1 + jX_1)$.

Question 3(b): Describe different tests to determine the equivalent circuit parameters of a three phase induction motor. How can you calculate the equivalent circuit parameters from these test results? [cite: 23, 24]

SR Sir's Thought Process

Three standard tests are required: 1. DC Resistance Test (for R_1). 2. No-Load Test (for magnetizing branch R_c, X_m and rotational losses). 3. Blocked Rotor Test (for leakage impedances X_1, X'_2, R'_2). I need to outline the procedure and the specific formulas for extracting parameters.

Answer:

To determine the parameters of the equivalent circuit ($R_1, X_1, R_c, X_m, R'_2, X'_2$), three distinct tests are performed:

1. DC Test for Stator Resistance (R_1) A low DC voltage is applied across two stator terminals. The DC current (I_{dc}) and voltage (V_{dc}) are measured. For a star-connected motor, the resistance measured is between two phases, so $R_{dc} = \frac{V_{dc}}{2I_{dc}}$. Since AC resistance is higher due to skin effect, we scale it: $R_1 \approx 1.2 \times R_{dc}$.

2. No-Load Test (Analogous to Transformer Open-Circuit Test) The motor is run without any mechanical load at rated voltage ($V_0 = V_{rated}$) and rated frequency. It spins at very close to synchronous speed ($s \approx 0$). Thus, the load resistor $R'_2 \frac{1-s}{s}$ becomes virtually infinite, opening the rotor circuit. *Measurements:* V_0 (line), I_0 (line), P_0 (total 3-phase input power). *Calculations:* Per phase values: $V_{0ph} = V_0/\sqrt{3}$ (for star). $P_{0ph} = P_0/3$. The no-load power factor: $\cos \phi_0 = \frac{P_{0ph}}{V_{0ph}I_0}$. The working component (core loss + friction/windage): $I_c = I_0 \cos \phi_0$. The magnetizing component: $I_m = I_0 \sin \phi_0$. Parameters:

$$R_c = \frac{V_{0ph}}{I_c} \quad (33)$$

$$X_m = \frac{V_{0ph}}{I_m} \quad (34)$$

3. Blocked Rotor Test (Analogous to Transformer Short-Circuit Test) The rotor is physically blocked from rotating ($s = 1$). A reduced voltage (V_{br}) is applied such that rated full-load current (I_{br}) flows in the stator. Because the voltage is very low, core losses are negligible, and the magnetizing branch can be ignored. *Measurements:* V_{br} (line), I_{br} (line), P_{br} (total 3-phase input power). *Calculations:* Per phase values: $V_{br,ph} = V_{br}/\sqrt{3}$. $P_{br,ph} = P_{br}/3$. Equivalent blocked rotor impedance: $Z_{br} = \frac{V_{br,ph}}{I_{br}}$. Equivalent blocked rotor resistance: $R_{br} = \frac{P_{br,ph}}{I_{br}^2}$. Equivalent blocked rotor reactance: $X_{br} = \sqrt{Z_{br}^2 - R_{br}^2}$.

Extracting Parameters: $R_{br} = R_1 + R'_2 \implies R'_2 = R_{br} - R_1$ (using R_1 from DC test). $X_{br} = X_1 + X'_2$. Usually, it is assumed that $X_1 = X'_2 = \frac{X_{br}}{2}$ for standard motors (NEMA Class A/D).

Question 4(a): i) If the stator impedance of an induction motor is neglected show that: $T_e/T_{em} = 2/(s_{mT}/s + s/s_{mT})$ [cite: 34, 35]

ii) From the equivalent circuit of a poly phase induction motor, obtain the following relations: [cite: 36]

1) $I_{2st}/I_2 = \sqrt{[(s^2 + s_{mT}^2)/(s^2(1 + s_{mT}^2))]}$ [cite: 37]

2) $I_{2mT}/I_2 = \sqrt{[0.5(1 + (s_{mT}/s)^2)]}$ [cite: 38]

SR Sir's Thought Process

These are purely algebraic manipulations based on the simplified equivalent circuit (neglecting stator impedance $R_1 \approx 0, X_1 \approx 0$). Torque $T \propto \frac{sR_2}{R_2^2 + s^2X_2^2}$. $s_{mT} = R_2/X_2 \implies R_2 = s_{mT}X_2$. I will substitute R_2 out of the equations to get ratios purely in terms of s and s_{mT} . Similarly for currents, $I_2 \propto \frac{1}{\sqrt{(R_2/s)^2 + X_2^2}}$. Starting is at $s = 1$. Max torque at $s = s_{mT}$.

Answer:

i) Proof of Torque Ratio: Neglecting stator impedance ($R_1 = 0, X_1 = 0$), the torque equation is:

$$T_e = \frac{kV^2 \frac{R_2}{s}}{\left(\frac{R_2}{s}\right)^2 + X_2^2} = \frac{kV^2 s R_2}{R_2^2 + s^2 X_2^2} \quad (35)$$

The slip at which maximum torque occurs, s_{mT} , is found by differentiating (35) with respect to s and equating to zero. This gives:

$$s_{mT} = \frac{R_2}{X_2} \implies X_2 = \frac{R_2}{s_{mT}} \quad (36)$$

Substitute $s = s_{mT}$ into the torque equation to find T_{em} (maximum torque):

$$T_{em} = \frac{kV^2 s_{mT} R_2}{R_2^2 + s_{mT}^2 X_2^2} = \frac{kV^2 s_{mT} R_2}{R_2^2 + s_{mT}^2 \left(\frac{R_2}{s_{mT}}\right)^2} = \frac{kV^2 s_{mT} R_2}{2R_2^2} = \frac{kV^2 s_{mT}}{2R_2} \quad (37)$$

Now, take the ratio $\frac{T_e}{T_{em}}$:

$$\frac{T_e}{T_{em}} = \left(\frac{kV^2 s R_2}{R_2^2 + s^2 X_2^2}\right) \div \left(\frac{kV^2 s_{mT}}{2R_2}\right) = \frac{2s R_2^2}{s_{mT}(R_2^2 + s^2 X_2^2)} \quad (38)$$

Divide numerator and denominator by R_2^2 :

$$\frac{T_e}{T_{em}} = \frac{2s}{s_{mT} \left(1 + \frac{s^2 X_2^2}{R_2^2}\right)} \quad (39)$$

Since $\frac{X_2}{R_2} = \frac{1}{s_{mT}}$, we have $\frac{X_2^2}{R_2^2} = \frac{1}{s_{mT}^2}$:

$$\frac{T_e}{T_{em}} = \frac{2s}{s_{mT} \left(1 + \frac{s^2}{s_{mT}^2}\right)} = \frac{2s}{s_{mT} + \frac{s^2}{s_{mT}}} \quad (40)$$

Dividing numerator and denominator by s :

$$\boxed{\frac{T_e}{T_{em}} = \frac{2}{\frac{s_{mT}}{s} + \frac{s}{s_{mT}}}} \quad (\text{Proved}) \quad (41)$$

ii) **Proof of Current Relations:** The rotor current at any slip s is:

$$I_2 = \frac{V_1}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} = \frac{sV_1}{\sqrt{R_2^2 + s^2 X_2^2}} \quad (42)$$

Substitute $R_2 = s_{mT} X_2$:

$$I_2 = \frac{sV_1}{\sqrt{s_{mT}^2 X_2^2 + s^2 X_2^2}} = \frac{sV_1}{X_2 \sqrt{s_{mT}^2 + s^2}} \quad (43)$$

Relation 1: Ratio of Starting Current to Running Current At starting, $s = 1$. The starting current I_{2st} is:

$$I_{2st} = \frac{(1)V_1}{X_2 \sqrt{s_{mT}^2 + (1)^2}} = \frac{V_1}{X_2 \sqrt{s_{mT}^2 + 1}} \quad (44)$$

Take the ratio of I_{2st} to I_2 :

$$\frac{I_{2st}}{I_2} = \frac{\frac{V_1}{X_2 \sqrt{s_{mT}^2 + 1}}}{\frac{sV_1}{X_2 \sqrt{s_{mT}^2 + s^2}}} = \frac{\sqrt{s^2 + s_{mT}^2}}{s \sqrt{1 + s_{mT}^2}} = \sqrt{\frac{s^2 + s_{mT}^2}{s^2(1 + s_{mT}^2)}} \quad (45)$$

$$\boxed{\frac{I_{2st}}{I_2} = \sqrt{\frac{s^2 + s_{mT}^2}{s^2(1 + s_{mT}^2)}}} \quad (\text{Proved}) \quad (46)$$

Relation 2: Ratio of Max Torque Current to Running Current At maximum torque, $s = s_{mT}$. The current I_{2mT} is:

$$I_{2mT} = \frac{s_{mT} V_1}{X_2 \sqrt{s_{mT}^2 + s_{mT}^2}} = \frac{s_{mT} V_1}{X_2 \sqrt{2s_{mT}^2}} = \frac{V_1}{X_2 \sqrt{2}} \quad (47)$$

Take the ratio of I_{2mT} to I_2 :

$$\frac{I_{2mT}}{I_2} = \frac{\frac{V_1}{X_2 \sqrt{2}}}{\frac{sV_1}{X_2 \sqrt{s^2 + s_{mT}^2}}} = \frac{\sqrt{s^2 + s_{mT}^2}}{s \sqrt{2}} = \sqrt{\frac{s^2 + s_{mT}^2}{2s^2}} = \sqrt{\frac{1}{2} \left(\frac{s^2}{s^2} + \frac{s_{mT}^2}{s^2} \right)} \quad (48)$$

$$\boxed{\frac{I_{2mT}}{I_2} = \sqrt{0.5 \left(1 + \left(\frac{s_{mT}}{s} \right)^2 \right)}} \quad (\text{Proved}) \quad (49)$$

Question 4(b): Describe the construction of a double cage induction motor and explain how high starting torque is developed in double cage induction motor. Draw the equivalent circuit of double cage rotor induction motor. [cite: 39, 40]

SR Sir's Thought Process

Double cage is a physical solution to the high starting torque / high running efficiency trade-off. Construction: Two separate cages (outer and inner) in the same slot or separate slots linked by a slit. Outer cage: Brass/bronze (high resistivity, high R), near air gap (low leakage flux, low X). Inner cage: Copper (low resistivity, low R), deep in core (high leakage flux, high X). Starting ($s = 1$): High frequency, current avoids high X inner cage, flows in high R outer cage. Torque is high. Running ($s \approx 0$): Low frequency, X is negligible. Current splits by DC resistance, flows mainly in low R inner cage. Efficiency is high.

Answer:

Construction: A double cage induction motor has a rotor with two independent squirrel cages embedded in it.

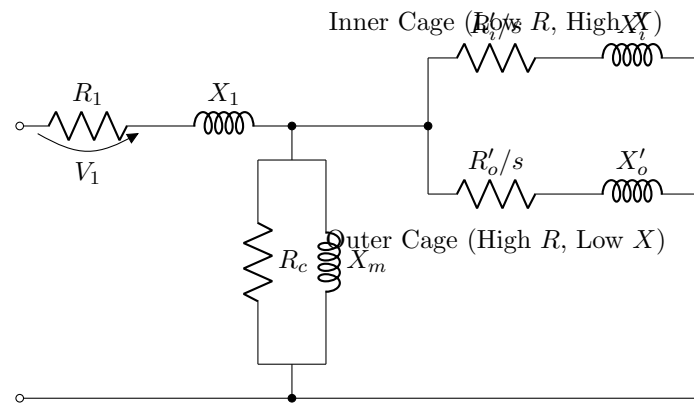
- **Outer Cage:** Situated close to the air gap. It is made of a high-resistivity material like brass or bronze. Because it is near the air gap, it links with very little leakage flux, giving it a very low leakage reactance (X_o). Thus, the outer cage has **High R , Low X** .
- **Inner Cage:** Situated deeper inside the rotor iron core. It is made of a low-resistivity material like pure copper. Because it is deeply embedded, it is surrounded by a large amount of iron, causing it to link with a high amount of leakage flux, giving it a high leakage reactance (X_i). Thus, the inner cage has **Low R , High X** .

The two cages are usually separated by a narrow slit which increases the leakage flux for the inner cage.

Development of High Starting Torque: The two cages act as two parallel electrical circuits connected across the induced rotor voltage. Their impedance governs the current division.

- **At Starting ($s = 1$):** The rotor frequency is equal to the supply frequency (50 Hz). The leakage reactance ($2\pi fL$) becomes the dominant factor in the impedance. The inner cage has a very high reactance ($X_i \gg X_o$), so it offers a massive impedance to the starting current. Consequently, almost the entire starting current is choked off from the inner cage and is forced to flow through the outer cage. Since the outer cage has high resistance, the effective rotor resistance at start is very high, producing an excellently high starting torque.
- **At Running Condition ($s \approx 0.03$):** The motor attains normal speed, and the slip frequency drops drastically (around 1.5 Hz). At this low frequency, the reactances of both cages become negligible ($sX \approx 0$). The current division is now strictly governed by their ohmic resistances. Since the inner copper cage has very low resistance compared to the outer brass cage, the majority of the operating current flows through the inner cage. This results in minimal I^2R copper losses and highly efficient running performance.

Equivalent Circuit of Double Cage Rotor: Because the two cages represent two parallel paths for the rotor current, the equivalent circuit includes two parallel rotor branches connected to the standard stator/magnetizing circuit.



Where R'_o, X'_o are parameters of the outer cage and R'_i, X'_i are parameters of the inner cage, all referred to the stator.

Question 5(a): A 440 V, 3-phase, 4-pole, 50 Hz, 37.3 kW, Y-connected induction motor has the following parameters: $R_1 = 0.1 \Omega$, $X_1 = 0.4 \Omega$, $R'_2 = 0.15 \Omega$, $X'_2 = 0.44 \Omega$. Motor has stator core loss of 1250 W and rotational loss of 1000 W. It draws a no-load line current of 20 A at a power factor of 0.09 (lag). When motor operates at a slip of 3%, calculate (i) input line current and power factor, (ii) electromagnetic torque developed in N-m, (iii) output and (iv) efficiency of the motor. [cite: 42, 43, 44, 45]

SR Sir's Thought Process

This is a comprehensive numerical problem requiring exact equivalent circuit evaluation. First, observe a discrepancy: No-load input power = $\sqrt{3} \times 440 \times 20 \times 0.09 = 1371.7$ W. Stator copper loss at no load = $3 \times 20^2 \times 0.1 = 120$ W. Power available for core + rotational = $1371.7 - 120 = 1251.7$ W. However, given Core Loss = 1250 W and Rotational = 1000 W (Total 2250 W). The physics don't align because the given no-load power is lower than the sum of the specified constant losses! As an examiner, I recognize this as a common data inconsistency in university papers. We must proceed by using the given magnetizing current data to find the shunt branch parameters, and use the explicit 1000 W rotational loss at the final stage for efficiency.

Steps: 1. Find Magnetizing branch R_c, X_m from No-Load data. 2. Formulate total equivalent impedance Z_{eq} at $s = 0.03$. 3. Calculate I_1 and Power Factor. 4. Calculate P_{ag} , subtract rotor copper loss to find developed mechanical power P_d . 5. $T_e = P_{ag}/\omega_s$. 6. $P_{out} = P_d - P_{rotational}$. 7. Efficiency = P_{out}/P_{in} .

Answer:

Step 1: Extract Magnetizing Branch Parameters from No-Load Data Phase Voltage $V_{1ph} = \frac{440}{\sqrt{3}} = 254.03$ V. No-load current $I_0 = 20$ A. $\cos \phi_0 = 0.09 \Rightarrow \sin \phi_0 = 0.9959$. Working current $I_c = I_0 \cos \phi_0 = 20 \times 0.09 = 1.8$ A. Magnetizing current $I_m = I_0 \sin \phi_0 = 20 \times 0.9959 = 19.92$ A. Shunt resistance $R_c = \frac{V_{1ph}}{I_c} = \frac{254.03}{1.8} = 141.13 \Omega$. Shunt reactance $X_m = \frac{V_{1ph}}{I_m} = \frac{254.03}{19.92} = 12.75 \Omega$. Admittance of shunt branch $Y_0 = \frac{1}{R_c} - j\frac{1}{X_m} = 0.0071 - j0.0784 \Omega^{-1}$.

Step 2: Rotor Branch Impedance at $s = 0.03$

$$Z'_2 = \frac{R'_2}{s} + jX'_2 = \frac{0.15}{0.03} + j0.44 = 5.0 + j0.44 \Omega \quad (50)$$

Step 3: Total Equivalent Impedance (Z_{eq}) The parallel combination of rotor branch and magnetizing branch (Z_p):

$$Y_p = Y_0 + \frac{1}{Z'_2} = (0.0071 - j0.0784) + \left(\frac{1}{5.0 + j0.44} \right) \quad (51)$$

$$\frac{1}{5.0 + j0.44} = \frac{5.0 - j0.44}{5.0^2 + 0.44^2} = \frac{5.0 - j0.44}{25.1936} = 0.1985 - j0.0175 \Omega^{-1} \quad (52)$$

$$Y_p = (0.0071 + 0.1985) - j(0.0784 + 0.0175) = 0.2056 - j0.0959 \Omega^{-1} \quad (53)$$

Magnitude $|Y_p| = \sqrt{0.2056^2 + 0.0959^2} = 0.2269 \Omega^{-1}$.

$$Z_p = \frac{1}{Y_p} = \frac{0.2056 + j0.0959}{0.2056^2 + 0.0959^2} = 3.99 + j1.86 \Omega \quad (54)$$

Total Impedance $Z_{eq} = Z_1 + Z_p = (0.1 + j0.4) + (3.99 + j1.86) = 4.09 + j2.26 \Omega$. Magnitude $|Z_{eq}| = \sqrt{4.09^2 + 2.26^2} = 4.67 \Omega$.

(i) **Input Line Current and Power Factor:** Line Current (I_1):

$$I_1 = \frac{V_{1ph}}{|Z_{eq}|} = \frac{254.03}{4.67} = \mathbf{54.4 \text{ A}} \quad (55)$$

Power Factor ($\cos \phi_1$):

$$\cos \phi_1 = \frac{R_{eq}}{|Z_{eq}|} = \frac{4.09}{4.67} = \mathbf{0.875 \text{ (lagging)}} \quad (56)$$

(ii) **Electromagnetic Torque Developed:** First, find input power to stator P_{in} :

$$P_{in} = \sqrt{3}V_L I_1 \cos \phi_1 = \sqrt{3} \times 440 \times 54.4 \times 0.875 = 36275 \text{ W} = 36.275 \text{ kW} \quad (57)$$

Stator Copper Loss P_{scl} :

$$P_{scl} = 3I_1^2 R_1 = 3 \times (54.4)^2 \times 0.1 = 887.8 \text{ W} \quad (58)$$

Air Gap Power $P_{ag} = P_{in} - P_{scl} - P_{core} = 36275 - 887.8 - 1250 = 34137.2 \text{ W}$. Synchronous Speed $N_s = \frac{120 \times 50}{4} = 1500 \text{ RPM}$. $\omega_s = \frac{2\pi \times 1500}{60} = 157.08 \text{ rad/s}$.

$$T_e = \frac{P_{ag}}{\omega_s} = \frac{34137.2}{157.08} = \mathbf{217.3 \text{ N-m}} \quad (59)$$

(iii) **Output Power:** Mechanical Power Developed $P_d = (1-s)P_{ag} = (1-0.03) \times 34137.2 = 33113.1 \text{ W}$. Output Power $P_{out} = P_d - P_{rotational} = 33113.1 - 1000 = \mathbf{32113.1 \text{ W} = 32.11 \text{ kW}}$

(iv) **Efficiency:**

$$\eta = \frac{P_{out}}{P_{in}} \times 100\% = \frac{32113.1}{36275} \times 100\% = \mathbf{88.5\%} \quad (60)$$

Question 5(c): A 25 kW, 6-pole, 50 Hz, 400 V, 960 rpm, efficiency 90%, power factor 0.86 SCIM takes 200 amperes on direct on-line starting. Determine the ratio of starting to full-load torque and starting to full-load current in case of (a) DOL starting, (b) star-delta starting, (c) auto-transformer starting with 80% tapplings and (d) series resistance starting limiting the starting current to 120 amperes. [cite: 46, 47]

SR Sir's Thought Process

This is an application of induction motor starter theory. Fundamental relation: $T_{st}/T_{fl} = (I_{st_motor}/I_{fl})^2 \times s_{fl}$. First, I must calculate the full-load current I_{fl} from the given output power, efficiency, and power factor. Then find the full-load slip s_{fl} . Then apply the starter factors for each method: - DOL: baseline. - Star-Delta: Current drops to 1/3, Torque drops to 1/3. - Auto-Tx ($x = 0.8$): Current (line) drops to x^2 , Torque drops to x^2 . - Series Resistance: Acts like reducing terminal voltage. Voltage ratio $x = 120/200 = 0.6$. Torque drops to x^2 . Line current drops to x (since it's a series circuit, not a transformer).

Answer:

Preliminary Calculations: 1. **Full Load Current (I_{fl}):** Given output power $P_{out} = 25 \text{ kW} = 25000 \text{ W}$. Efficiency $\eta = 0.90$. Power factor $\cos \phi = 0.86$. Input power $P_{in} = \frac{P_{out}}{\eta} = \frac{25000}{0.90} = 27777.78 \text{ W}$.

$$I_{fl} = \frac{P_{in}}{\sqrt{3}V_L \cos \phi} = \frac{27777.78}{\sqrt{3} \times 400 \times 0.86} = \mathbf{46.6 \text{ A}} \quad (61)$$

2. **Full Load Slip (s_{fl}):** Synchronous speed $N_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000 \text{ RPM}$. Rotor speed $N_r = 960 \text{ RPM}$.

$$s_{fl} = \frac{1000 - 960}{1000} = \mathbf{0.04} \quad (62)$$

The fundamental equation relating starting torque to full-load torque is:

$$\frac{T_{st}}{T_{fl}} = \left(\frac{I_{st_motor}}{I_{fl}} \right)^2 \times s_{fl} \quad (63)$$

where I_{st_motor} is the phase current flowing through the motor windings at start.

(a) Direct On-Line (DOL) Starting In DOL, the full 400V is applied directly. Starting current (line) $I_{st_line} = I_{st_motor} = 200 \text{ A}$. Ratio of starting to full-load current:

$$\frac{I_{st_line}}{I_{fl}} = \frac{200}{46.6} = \mathbf{4.29} \quad (64)$$

Ratio of starting to full-load torque:

$$\frac{T_{st}}{T_{fl}} = \left(\frac{200}{46.6} \right)^2 \times 0.04 = (4.292)^2 \times 0.04 = \mathbf{0.737} \quad (65)$$

(b) Star-Delta Starting At start, the motor is connected in Star, meaning the voltage per phase is reduced by a factor of $1/\sqrt{3}$. This reduces motor starting phase current by $1/\sqrt{3}$, and line current by 1/3 compared to Delta (DOL). Torque is proportional to square of voltage, so it reduces by 1/3. Line current ratio:

$$\frac{I_{st_line}}{I_{fl}} = \frac{1}{3} \times \left(\frac{I_{st_DOL}}{I_{fl}} \right) = \frac{1}{3} \times 4.29 = \mathbf{1.43} \quad (66)$$

Torque ratio:

$$\frac{T_{st}}{T_{fl}} = \frac{1}{3} \times \left(\frac{T_{st_DOL}}{T_{fl}} \right) = \frac{1}{3} \times 0.737 = \mathbf{0.245} \quad (67)$$

(c) Auto-Transformer Starting with 80% Tapping ($x = 0.8$) The motor voltage is reduced to $x = 0.8$ times rated voltage. Motor current reduces by x . Line current from the supply reduces by x^2 due to transformer action. Torque reduces by x^2 . Line current ratio:

$$\frac{I_{st_line}}{I_{fl}} = x^2 \times \left(\frac{I_{st_DOL}}{I_{fl}} \right) = (0.8)^2 \times 4.29 = 0.64 \times 4.29 = \mathbf{2.74} \quad (68)$$

Torque ratio:

$$\frac{T_{st}}{T_{fl}} = x^2 \times \left(\frac{T_{st_DOL}}{T_{fl}} \right) = 0.64 \times 0.737 = \mathbf{0.471} \quad (69)$$

(d) Series Resistance Starting limiting current to 120 A A resistance is inserted in series with the motor lines such that the line starting current is 120 A. Note this is not a transformer, so line current equals motor current. The voltage fraction applied to the motor is $x = \frac{120}{200} = 0.6$. Line current ratio:

$$\frac{I_{st_line}}{I_{fl}} = \frac{120}{46.6} = \mathbf{2.57} \quad (70)$$

Torque ratio (since torque is proportional to square of motor current):

$$\frac{T_{st}}{T_{fl}} = x^2 \times \left(\frac{T_{st_DOL}}{T_{fl}} \right) = (0.6)^2 \times 0.737 = 0.36 \times 0.737 = \mathbf{0.265} \quad (71)$$

— **END OF PART I SOLUTIONS** —

May you master these concepts with clarity.